

Workshop Activities

Activity 1 – Frequency Distributions for Categorical Data

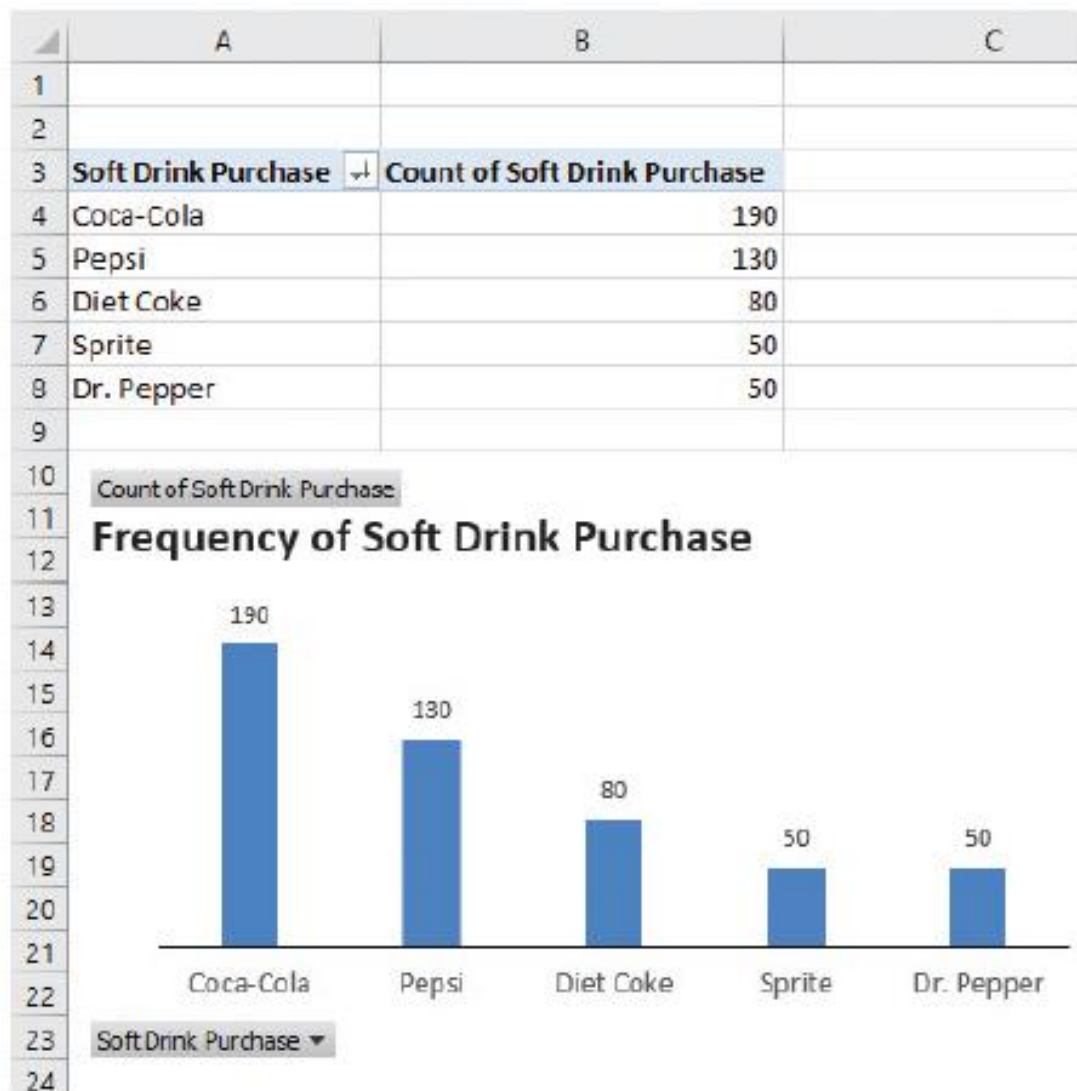
Create a column chart for the categorical data in the *Pop* data file. The file *Pop* contains data on a sample of 500 soft drinks. A portion of the data is shown below.

	A
1	Soft Drink Purchase
2	Coca-Cola
3	Diet Coke
4	Pepsi
5	Diet Coke
6	Coca-Cola
7	Coca-Cola
8	Dr. Pepper
9	Diet Coke
10	Pepsi
11	Pepsi

Open the *Pop* data file and complete the following steps:

- Step 1.
 - Select cells A1:A501
- Step 2.
 - Click the Insert tab on the Ribbon
- Step 3.
 - Click the Recommended Charts button in the Charts group
- Step 4.
 - When the Insert Chart dialog box appears:
 - Select Clustered Column
 - Click OK
- Step 5.
 - Click any of the columns in the chart that appears. While the columns are selected, right click a column, then select Sort and Sort Largest to Smallest

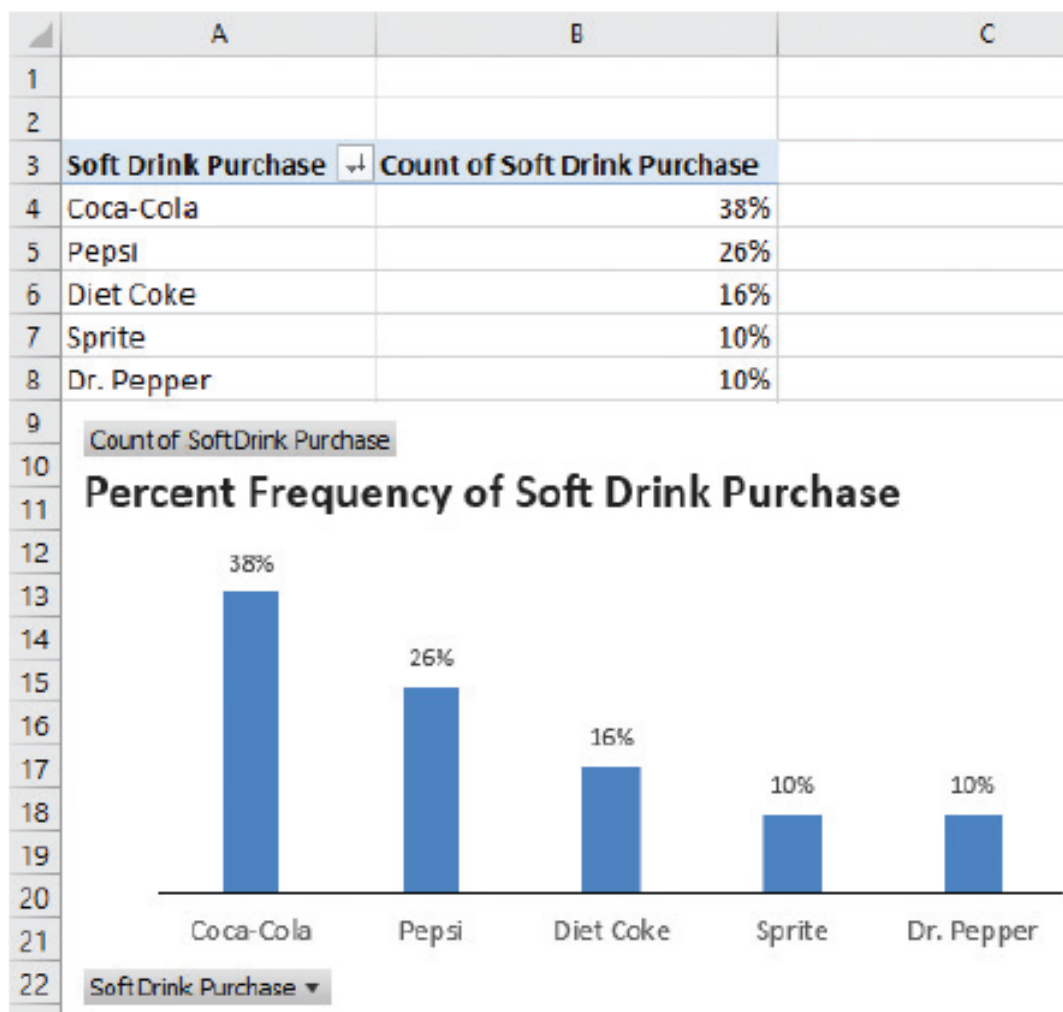
The result should look similar to the following pivot table and chart.



A relative frequency can be expressed as a percentage. To obtain the percent frequency distribution, complete the following steps:

- Step 1.
 - Select any cell in the Count of Soft Drink Purchase column of the PivotTable (any cell in range B3:B8)
- Step 2.
 - When the PivotTable Fields task pane appears:
 - In the Values area, select the triangle to the right of Count of Soft Drink Purchase
 - Select Value Field Settings from the list of options.
- Step 3.
 - When the Value Field Settings dialog box appears:
 - Click the Show Values As tab and in the box below Show values as select % of Grand Total

The result should look similar to the following pivot table and chart.



Which of these two charts might be considered to be more useful? Justify your choice.

Activity 2 – Creating a Histogram

Create a histogram for the quantitative data in the *Death* data file. The file *Death* contains a data set that contains the age at death for 700 individuals. A portion of the data is shown below.

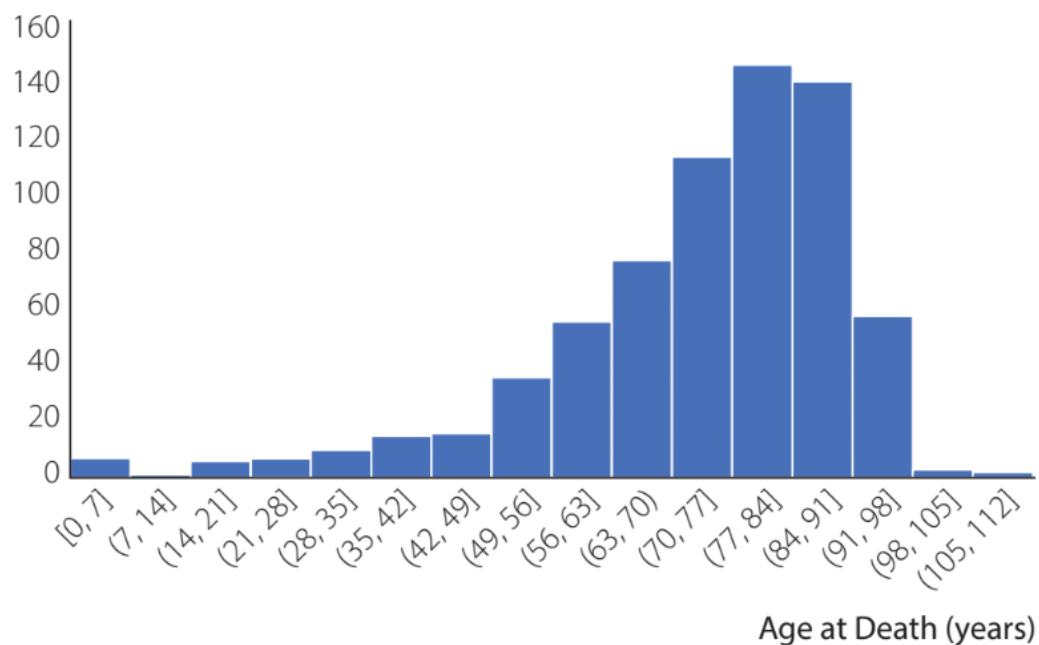
	A
1	Age at Death (Years)
2	83
3	76
4	78
5	74
6	35
7	78
8	73
9	84
10	55
11	73

Open the *Death* data file and complete the following steps:

- Step 1.
 - Select cells A1:A701
- Step 2.
 - Click the Insert tab on the Ribbon
- Step 3.
 - Click the Insert Statistic Chart button in the Charts group
 - When the list of statistic charts appears, select Histogram

Edit the chart to improve its appearance (see Workshop 1, Activity 5). The result should look similar to the following histogram.

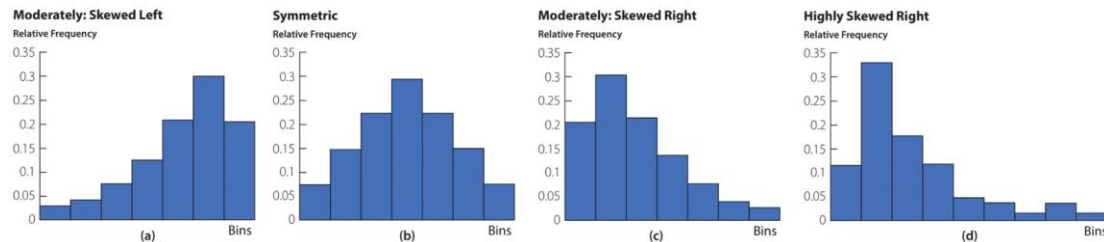
Distribution of Age at Death



Discuss the insights that can be drawn from this histogram.

Activity 3 – Skewness

Consider the four examples of histograms with different levels of skewness shown below.



Using your common sense and your general knowledge, try to work out the expected skewness (either left, symmetric, or right) for the following examples and in each case explain your logic:

1. Age of death
2. Scores in easy exam
3. Housing prices
4. Adult male height

Activity 4 – The Frequency Polygon

Like a histogram, a frequency polygon plots frequency counts of observations in a set of bins. However, it uses lines instead of columns to connect the counts of different bins. In this activity you will create a frequency polygon using data from the *DeathTwo* data file. The *DeathTwo* data file contains a data set that contains the age at death for 700 individuals.

Open the *DeathTwo* data file and complete the following steps to create frequency distributions to build histograms and frequency polygons for the variables 'Male' and 'Female'.

- Step 1. Compute the number of Female and Male observations:
 - Cell F3: enter =COUNTIF(B2:B701,"Female")
 - Cell G3: enter =COUNTIF(B2:B701,"Male")
- Step 2. Enter the bins lower and upper limits.
 - Cell D5: enter "Bin Lower Limit"
 - Cell E5: enter "Bin Upper Limit"
 - Range D6:D21: enter the values 0, 7, 14, ... , 105
 - Range E6:E21: enter the values 7, 14, 21, ... , 112
- Step 3. Sort the data set by Sex, so that female data are from rows 2 to 328, and male data from rows 329 to 701.
- Step 4. Create the relative frequencies for the bins of males and females.

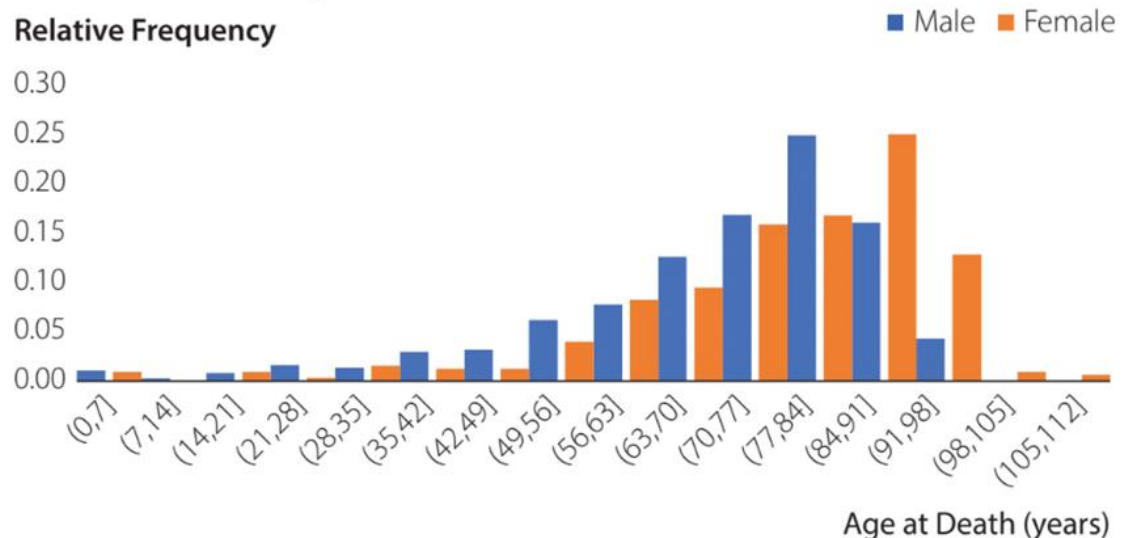
- Cell F5: enter "Female"
- Cell G5: enter "Male"
- Cell F6: enter =FREQUENCY(A2:A328,E6:E21)/\$F\$3
- Cell G6: enter =FREQUENCY(A329:A701,E6:E21)/\$G\$3
- FREQUENCY is a dynamic array formula. When you hit enter for the formula in F6 for the female bins, it will populate the cell range F6:F22, and when you hit enter for the formula in G6 for the male bins, it will populate the range G6:G22.
- Step 5. Create the labels for the horizontal axis.
 - Cell H5: enter "Labels"
 - Cell H6: enter =CONCAT("(",D6,"",E6,""]")
 - Copy cell H6 and paste it to cells H7:H21

The following steps produce a chart with histograms for the male and female observations.

- Step 1.
 - Select cells F6:G21
- Step 2.
 - Click the Insert tab on the Ribbon
- Step 3.
 - Click the Insert Column or Bar Chart button in the Charts group
- Step 4.
 - When the list of column and bar chart subtypes appears, click the Clustered Column button

Edit the chart to improve its appearance (see Workshop 1, Activity 5). The result should look similar to the following clustered column chart.

Distribution of Age at Death



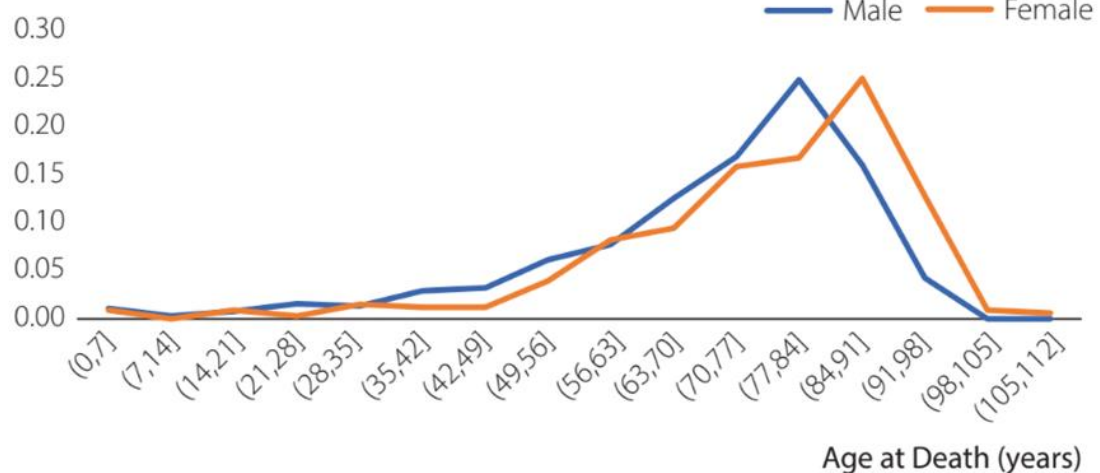
To display the age at death distributions for female and male observations as a frequency polygon, complete the following steps:

- Step 1.
 - Select cells F6:G21
- Step 2.
 - Click the Insert tab on the Ribbon
- Step 3.
 - Click the Insert Line or Area Chart button in the Charts group
 - When the list of line and area chart subtypes appears, click the Line button

Edit the chart to improve its appearance (see Workshop 1, Activity 5). The result should look similar to the following frequency polygon.

Distribution of Age at Death

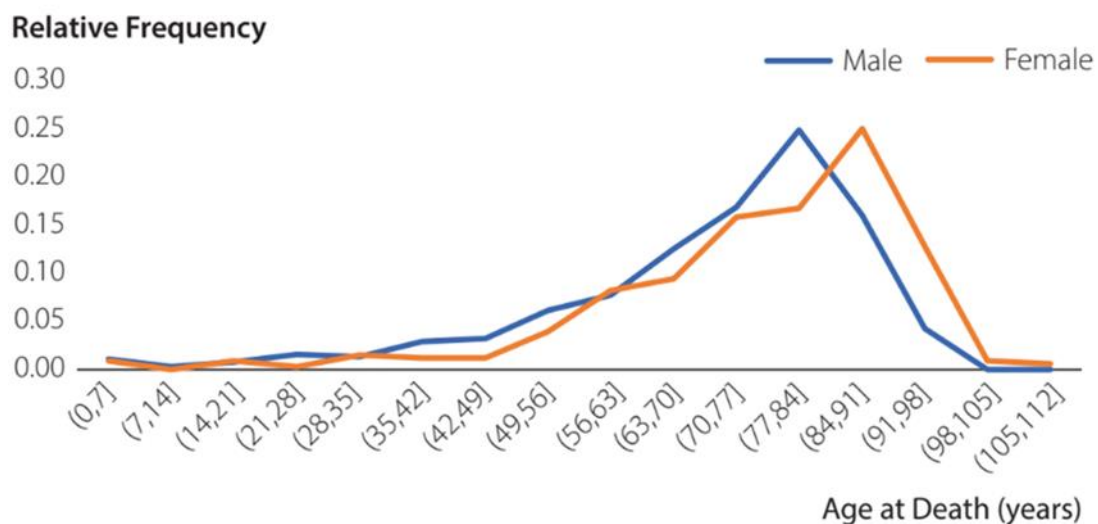
Relative Frequency



Activity 5 – Frequency Polygon considerations

Consider the frequency polygon you created in the previous activity.

Distribution of Age at Death



The distribution of female and male age at death is skewed left.

1. How could the summary statistics of location, such as mean and median, and the measure of variation, such as the quartiles, be used to predict the existence of skewness in the distributions of female and male age at death?
2. Which other data visualisation technique could be used to display the distribution of the female and male age at death, data?
3. How many outliers does each distribution have?

Activity 6 – Creating a Strip Chart and a Jittered Strip Chart

This activity creates a strip chart using data from the *HalfMarathon* data file. The *HalfMarathon* data file contains times for a collection of runners in a competitive half-marathon race. A sample of the data is shown below.

	A	B
1	Sex	Time (Minutes)
2	Male	148.70
3	Female	122.62
4	Male	127.98
5	Female	122.48
6	Female	111.22
7	Male	108.18
8	Female	189.27
9	Male	128.40
10	Female	153.88
11	Female	121.25

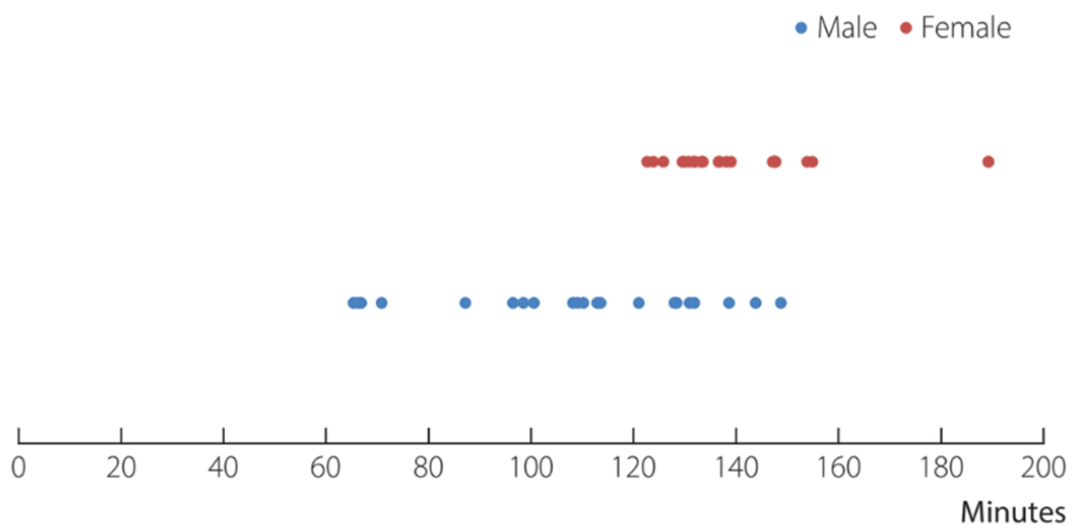
Open the *HalfMarathon* data file and complete the following steps:

- Step 1.
 - Select cells A1:B54
- Step 2.
 - Click the Data tab on the Ribbon
- Step 3.
 - Click the Sort button in the Sort & Filter group
- Step 4.
 - When the Sort dialog box appears, select the check box for My data has headers
 - In the Sort by row, select Sex for the Column entry, Cell Values for Sort on entry, and Z to A for Order entry
 - Click Add Level
 - In the Then by row, select Time (Minutes) for the Column entry, Cell Values for Sort on entry, and Smallest to Largest for Order entry
 - Click OK
- Step 5.
 - For each cell in the range C2:C23 (each cell corresponding to a Male finishing time), enter the value 10
- Step 6.
 - For each cell in the range C24:C54 (each cell corresponding to a Female finishing time), enter the value 20
- Step 7.
 - Select cells B2:C23 (cells corresponding to Male finishing times)
- Step 8.
 - Click the Insert tab on the Ribbon
- Step 9.

- Click the Insert Scatter (X,Y) or Bubble Chart button in the Charts group
- Step 10.
 - When the list chart subtypes appears, click the Scatter button
 - Right click the chart and select Select Data
- Step 11.
 - When the Select Data Source dialog box appears:
 - Select Series1 under Legend Entries (Series)
 - Click the Edit button
- Step 12.
 - When the Edit Series dialog box appears:
 - Enter Male in the box under Series name:
 - Click OK to close the Edit Series dialog box
- Step 13.
 - In the Select Data Source dialog box, click the Add button
- Step 14.
 - When the Edit Series dialog box appears:
 - Enter Female in the box under Series name:
 - Enter =Data!\$B\$24:\$B\$54 in the box under Series X values:
 - Enter =Data!\$C\$24:\$C\$54 in the box under Series Y values:
 - Click OK to close the Edit Series dialog box
 - Click OK to close the Select Data Source dialog box

Edit the chart to improve its appearance (see Workshop 1, Activity 5). The result should look similar to the following strip chart.

Half-Marathon Race Times



From the above strip chart the fastest and slowest times for females and for males, can be seen. However, the relative density of half-marathon times over the range is unclear. The issue is referred to as occlusion. To overcome occlusion, we can jitter the values.

To jitter the values, continue working on the strip chart by completing the following steps:

- Step 15.
 - Right click the Female data series in the chart and select Format Data Series
- Step 16.
 - When the Format Data Series task pane appears:
 - Click the Fill & Line button
 - Click Marker Fill
 - Under Fill, select No fill
- Step 17.
 - Leaving the Format Data Series task pane open, click the Male data series in the chart
 - Click the Fill & Line button
 - Click Marker Fill
 - Under Fill, select No fill

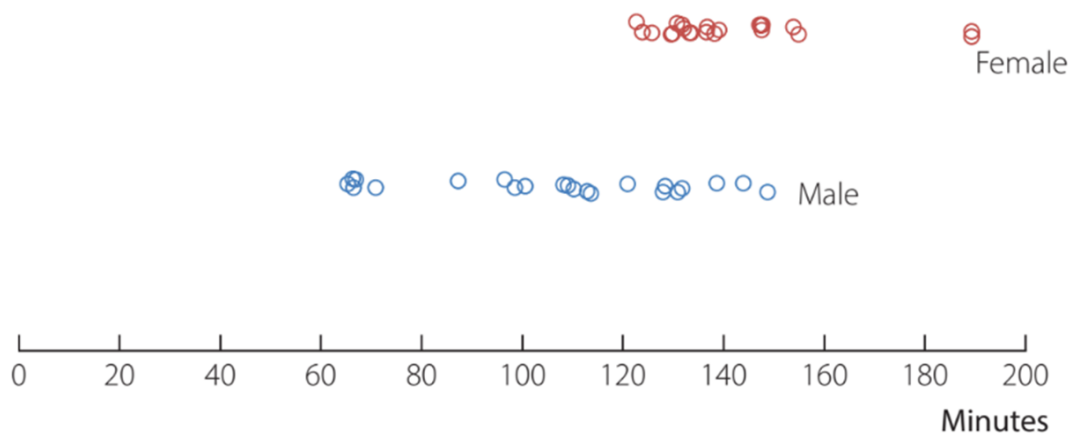
This results in the dots appearing hollow. However, there is still an element of occlusion. Continuing with the following steps enables the values to be jittered vertically producing a clearer display of the values.

- Step 18.
 - In cell D2, enter the formula =C2+RAND()
 - Copy the formula in cell D2 to cells D3:D54
- Step 19.
 - Right click the chart and choose Select Data...
- Step 20.
 - When the Select Data Source dialog box appears:
 - In the Legend Entries (Series) area, click Male and then click Edit
- Step 21.
 - When the Edit Series dialog box, appears:
 - Enter =Data!\$D\$2:\$D\$23 in the box under Series Y values:
 - Click OK to close the Edit Series dialog box
- Step 22.
 - In the Select Data Source dialog box:

- In the Legend Entries (Series) area, click Female and then click Edit
- Step 23.
 - When the Edit Series dialog box appears:
 - Enter =Data!\$D\$24:\$D\$54 in the box under Series Y values:
 - Click OK to close the Edit Series dialog box
 - Click OK to close the Select Data Source dialog box
- Step 24
 - Remove the Legend and add text boxes to label each jittered strip

The result should look similar to the following jittered strip chart.

Half-Marathon Race Times



Activity 7 – CEO Time in Meetings

A study was conducted to examine how chief executive officers (CEOs) spend their workdays. The file *CEOtime* contains a sample of data from this study, which tracked the minutes per day various CEOs spent in meetings.

- a. Construct a strip chart displaying the data making sure to employ techniques to avoid occlusion.
- b. What is the median amount of time a CEO spends in meetings per day?
- c. What is the mean amount of time a CEO spends in meetings per day?
- d. What is the most common amount of time a CEO spends in meetings per day?

Activity 8 – Creating a Box and Whisker Chart

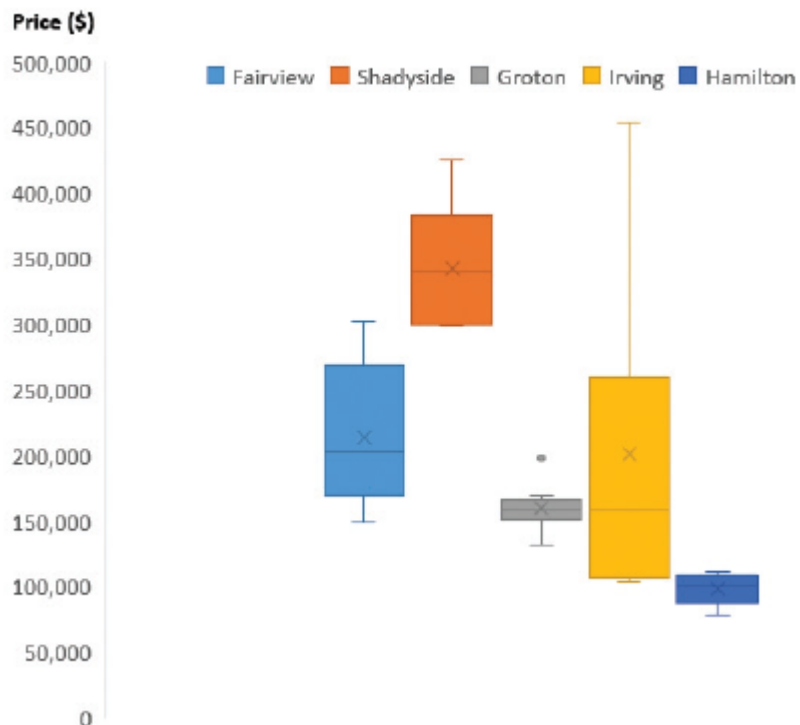
This activity creates a box and whisker chart using data from the *SalesComparison* data file. The *SalesComparison* data file compares home selling prices in five different suburbs.

Open the *SalesComparison* data file and complete the following steps:

- Step 1.
 - Select cells B1:F11
- Step 2.
 - Click the Insert tab on the Ribbon
- Step 3.
 - Click the Insert Statistic Chart button in the Charts group
 - When the list of statistic charts appears, select Box and Whisker

Edit the chart to improve its appearance, the result should look similar to the following box and whisker chart.

Comparing Home Selling Prices in Different Suburbs



What insights can be drawn from this box and whisker chart?

Activity 9 – Television Viewership of Football Games

Companies considering purchasing ads during televised XFL football games are interested in the viewership of games as a way of estimating the exposure their ads get. The file *PrimeTimeXFL* contains a sample of viewership numbers for XFL games televised on Friday and Saturday nights, respectively.

1. Construct box and whisker charts on the same chart that compare the viewership distribution of Friday night XFL games to the viewership distribution of Saturday night XFL games.
2. Use the box and whisker charts to compare the two distributions.
3. Use the box and whisker chart of the Friday night XFL games to describe this distribution.
4. Use the box and whisker chart of the Saturday night XFL games to describe this distribution.

Activity 10 – Column Charts with Confidence Interval Bars

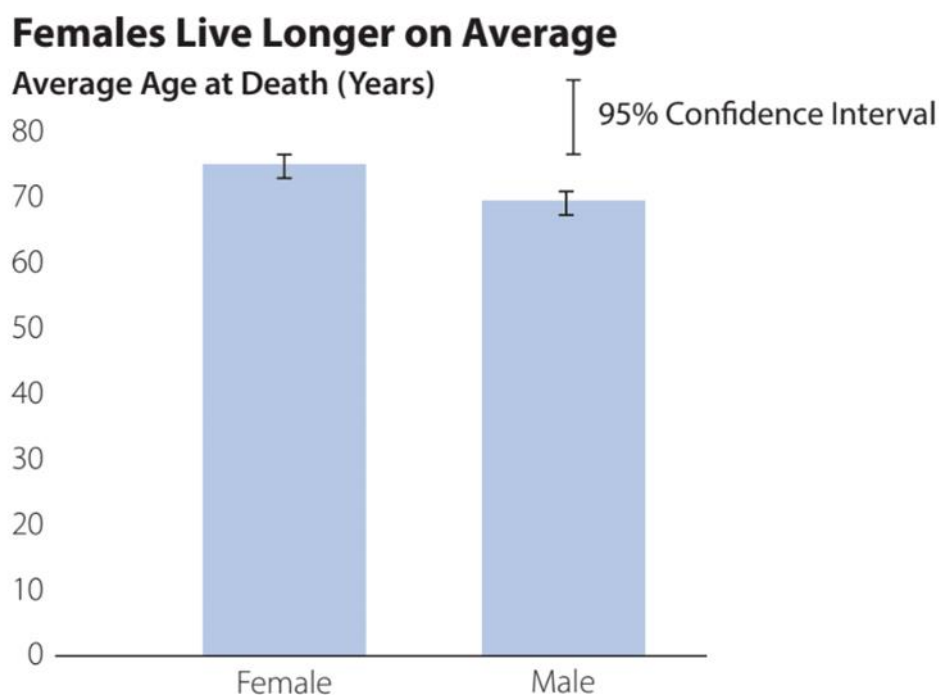
The purpose of a confidence interval is to provide information about how close the sample mean may be to the value of the population mean. A confidence interval is computed by adding and subtracting the margin of error. After calculation it can be displayed on a data visualisation.

To undertake this activity and visualise a confidence interval on a mean, open the *DeathAvgAgeChart* data file and complete the following steps:

- Step 1.
 - Click the column chart
- Step 2.
 - Click the Chart Elements button and select Error Bars
 - Click the black triangle to the right of Error Bars and select More Options...
- Step 3.
 - When the Format Error Bars task pane appears:
 - Click Error Bar Options
 - In the Error Amount area, select Custom
 - Click the Specify Value button to the right of Custom

- In the Custom Error Bars dialog box, enter `=Data!$E$5:$F$5` in both the Positive Error Value box and Negative Error Value box
- Step 4.
 - Manually add a text box to include a legend labelling the error bars as a 95% confidence interval. It is not necessary to include the error bar symbol.

Edit the chart to improve its appearance, the result should look similar to the following chart. Note: By right clicking the vertical axis and selecting Format Axis and then Axis Options, you can edit the minimum value of the bound on the vertical axis.



Activity 11 – Average Tax Payment Error

A sample of 10,001 Federal income tax returns with errors has been collected and the amount of payment error for each is provided in the file *FedTaxErrors*. A positive value indicates the taxpayer underpaid and a negative value indicates that the taxpayer overpaid.

1. What is the sample mean tax payment error made on these erroneous Federal income tax payments?
2. What is the sample standard deviation in payment error amount?

3. What is the margin of error for a 95% confidence interval on the mean payment error amount?
4. Construct a column chart displaying the sample mean tax payment error with error bars depicting a 95% confidence interval.
5. Can we conclude with 95% confidence that the mean tax payment error is positive (a taxpayer underpayment) or not? Explain your conclusion.

Activity 12 – Prediction Intervals for Regression

This activity creates a scatter chart display of observations, predictions and prediction interval limits using data from the *YourierChart* data file. Yourier LLC is a home delivery service that picks up items from stores and delivers them to customers. A sample of the data from the *YourierChart* data file is shown below.

	A	B	C	D	E	F
1	Route	Requests	Travel Time	Prediction	Lower 95% P.I.	Upper 95% P.I.
2		0		1.356	0.336	2.376
3	1	1	1.4	1.641	0.706	2.577
4	2	2	2.2	1.926	1.047	2.806
5	3	2	2.1	1.926	1.047	2.806
6	4	2	2	1.926	1.047	2.806
7	5	3	1.6	2.211	1.354	3.069
8	6	3	2.5	2.211	1.354	3.069
9	7	3	2.5	2.211	1.354	3.069
10	8	4	2	2.497	1.625	3.369
11	9	5	3	2.782	1.860	3.704
12	10	6	3.1	3.067	2.065	4.069
13		7		3.352	2.247	4.457

Using the data on the point predictions and the lower and upper limits of the 95% prediction intervals, the following steps demonstrate how to visualise the prediction information on a scatter chart of travel time versus number of requests.

Open the *YourierChart* data file:

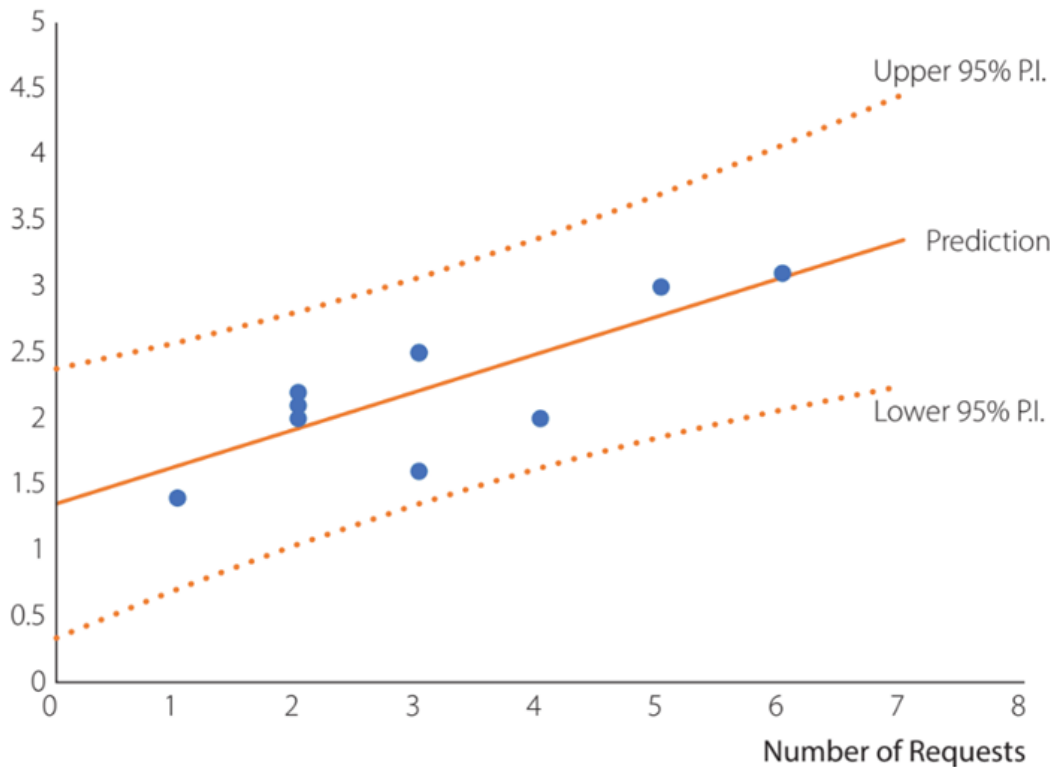
- Step 1.
 - Right click the chart and select Select Data
- Step 2.
 - When Select Data Source dialog box appears, click the Add button
- Step 3.
 - In the Edit Series dialog box:
 - Enter =Data!\$D\$1 in the box under Series name:
 - Enter =Data!\$B\$2:\$B\$13 in the box under Series X values:

- Enter =Data!\$D\$2:\$D\$13 in the box under Series Y values:
 - Click OK
- Step 4.
 - In the Select Data Source dialog box, click the Add button
- Step 5.
 - In the Edit Series dialog box:
 - Enter =Data!\$E\$1 in the box under Series name:
 - Enter =Data!\$B\$2:\$B\$13 in the box under Series X values:
 - Enter =Data!\$E\$2:\$E\$13 in the box under Series Y values:
 - Click OK
- Step 6.
 - In the Select Data Source dialog box, click the Add button
- Step 7.
 - In the Edit Series dialog box:
 - Enter =Data!\$F\$1 in the box under Series name:
 - Enter =Data!\$B\$2:\$B\$13 in the box under Series X values:
 - Enter =Data!\$F\$2:\$F\$13 in the box under Series Y values:
 - Click OK
 - Click OK to close Select Data Source dialog box
- Step 8.
 - Click a data series corresponding to either the point predictions or the prediction interval limits. With this data series selected, right-click, and then select Change Series Chart Type from the menu
- Step 9.
 - When the Change Chart Type dialog box appears, click X Y Scatter
 - Select Scatter with Smooth Lines on the All Charts tab
 - Click OK
- Step 10.
 - Click the blue line representing the observed travel times. Right click this line and select Format Data Series
- Step 11.
 - In the Format Data Series task pane, click the Fill & Line icon
 - Click Line and select No Line
 - Click Marker and under Marker Options select Automatic
- Step 12.
 - Select the chart and click the Chart Elements button and select Legend
 - Click the black triangle to the right of Legend and select Top

Edit the chart to improve its appearance, the result should look similar to the following chart.

Yourier LLC Route Performance

Travel Time (hours)



Discuss the conclusions that can be drawn from this regression model?

Activity 13 – Prediction Intervals for a Time Series Model

This activity creates a line chart to illustrate predictions on a time series chart using data from the *BundabergChart* data file. Bundaberg Brewed Drinks is a company that makes craft brewed sodas. Premium Australian root beer is an integral part of its product portfolio and Bundaberg Brewed Drinks is interested in forecasting its sales in future quarters. A sample of the data from the *BundabergChart* data file is shown below.

	A	B	C	D	E
1	Quarter	Sales	Prediction	Lower 95% P.I.	Upper 95% P.I.
2	1	86			
3	2	105			
4	3	191			
5	4	127			
6	5	83			
7	6	94			
8	7	173			
9	8	120			
10	9	90			
11	10	110			
12	11	188			
13	12	115			
14	13	98			
15	14	119			
16	15	188			
17	16	114			
18	17	74			
19	18		119.0	81.1	156.9
20	19		188.0	150.1	225.9
21	20		114.0	76.1	151.9
22	21		74.0	36.1	111.9
23	22		119.0	65.4	172.6
24	23		188.0	134.4	241.6
25	24		114.0	60.4	167.6
26	25		74.0	20.4	127.6

Using the data on the point predictions (in column C) and the lower and upper limits of the 95% prediction intervals (in columns D and E), the following steps demonstrate how to visualise this prediction information on the time series chart of quarterly root beer sales.

Open the *BundabergChart* data file:

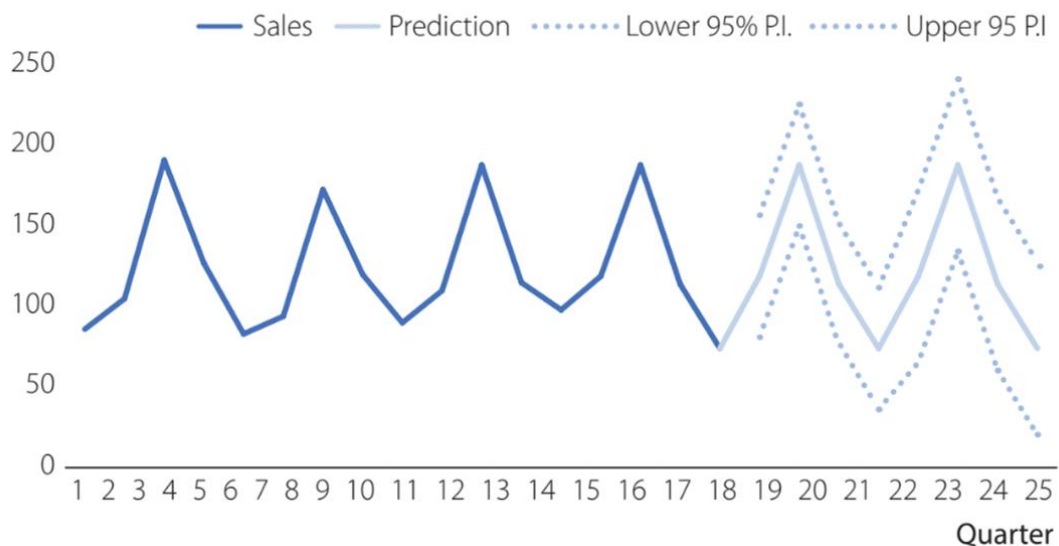
- Step 1.
 - In cell C18, enter =B18
- Step 2.
 - Right click the chart and select Select Data
- Step 3.
 - When Select Data Source dialog box appears, click the Add button
- Step 4.
 - In the Edit Series dialog box:
 - Enter =Data!\$C\$1 in the box under Series name:
 - Enter =Data!\$C\$2:\$C\$26 in the box under Series values:
 - Click OK
- Step 5.
 - In the Select Data Source dialog box, click the Add button
- Step 6.
 - In the Edit Series dialog box:
 - Enter =Data!\$D\$1 in the box under Series name:
 - Enter =Data!\$D\$2:\$D\$26 in the box under Series values:
 - Click OK
- Step 7.
 - In the Select Data Source dialog box, click the Add button

- Step 8.
 - In the Edit Series dialog box:
 - Enter =Data!\$E\$1 in the box under Series name:
 - Click OK to close Edit Series dialog box
 - Enter =Data!\$E\$2:\$E\$26 in the box under Series values:
 - Click OK
 - Click OK to close Select Data Source dialog box
- Step 9.
 - Select the chart and click the Chart Elements button, select Legend
 - Click the black triangle to the right of Legend, and select Top

Edit the chart to improve its appearance, the result should look similar to the following chart.

Bundaberg Root Beer Sales

Sales (\$1000s)



Discuss the conclusions that can be drawn from this time series model.